

Lab 21: K-Nearest Neighbors (KNN) Classification

Aim

To implement the **K-Nearest Neighbors (KNN) algorithm** for classification using the Iris dataset.

Algorithm

1. Import the required libraries.
2. Load the Iris dataset.
3. Split the dataset into training and testing sets.
4. Create the KNN classifier with a suitable value of K.
5. Train the model using the training data.
6. Predict the classes for the test data.
7. Calculate and display the accuracy.

Program

```
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from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score

# Load the Iris dataset
iris = load_iris()

X = iris.data
y = iris.target

# Split the dataset
```

```
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.2, random_state=42  
)
```

```
# Create KNN classifier
```

```
model = KNeighborsClassifier(n_neighbors=3)
```

```
# Train the model
```

```
model.fit(X_train, y_train)
```

```
# Make predictions
```

```
y_pred = model.predict(X_test)
```

```
print("Predictions:", y_pred)
```

```
print("Actual Values:", y_test)
```

```
print("Accuracy:", accuracy_score(y_test, y_pred))
```

Sample Output

```
Predictions: [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 1 0 2 2 2 2 2 0 0]
```

```
Actual Values: [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 1 0 2 2 2 2 2 0 0]
```

```
Accuracy: 1.0
```

Result

Thus, the **K-Nearest Neighbors (KNN) algorithm was successfully implemented** using the Iris dataset, and the data was classified with high accuracy.